

# AI characteristics and competitive advantage: the moderating role of resource allocation

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## Abstract

**Purpose** – This research explores how artificial intelligence's (AI's) distinct capabilities (interactivity, autonomy, inscrutability and abstraction), manifested as unique characteristics, impact decision-making in resource-limited social entrepreneurship, assessing their effect on competitive advantage with resource allocation as a key moderator. By analyzing such tensions, this research aims to bridge critical gaps in understanding how emerging technologies influence decisions in social entrepreneurship.

**Design/methodology/approach** – By adopting a dual theoretical framework (next-generation perceived characteristics of innovations [PCI] and resource-based view), this research employs a quantitative empirical approach by gathering data through e-surveys ( $n = 269$ ) from a professional database and two prominent conferences in AI and social entrepreneurship.

**Findings** – Linear and nonlinear relationships among AI characteristics and decision-making emerge, with the potential for moderation effects influenced by resource allocation.

**Originality/value** – This research makes four key contributions: empirically examining how distinct AI capabilities, manifested through unique characteristics, influence decision-making in the social entrepreneurship context; conceptually introducing “abstraction” as a novel AI capability; theoretically integrating the next-generation PCI framework with the resource-based view for a novel theoretical lens and practically developing a calibration graph as a “prototype” tool to quantify AI abstraction for resource-limited social entrepreneurship, thus potentially enabling optimal decision-making and consequently competitive advantage.

**Keywords** AI, Social entrepreneurship, Next-generation PCI, Resource-based view, Abstraction

**Paper type** Research article

## 1. Introduction

Digital technologies, particularly artificial intelligence (AI), have become central to reshaping organizational strategies across sectors (Aldoseri *et al.*, 2024). AI's capacity to process vast datasets, predict trends, and automate tasks has made it a catalyst for efficiency and innovation, transforming areas such as supply chain optimization and customer engagement (Kulkarni and Bansal, 2024). Unlike traditional technologies, AI introduces distinct capabilities, manifested as unique characteristics (e.g. interactivity, autonomy, and inscrutability), that redefine how systems operate and decisions are made. However, despite its disruptive potential, the systematic implications of AI for strategic and managerial decision-making remain underexplored and scarce (e.g. Neiroukh *et al.*, 2024), especially in resource-constrained and mission-driven contexts.

This gap is particularly salient in social entrepreneurship, where organizations operate under inherent uncertainty and face unique challenges in adopting AI (Townsend *et al.*, 2023). Due to limited resources, including capital, technology capacity, and workforce (Kamaludin *et al.*, 2024; Usman *et al.*, 2024; Gupta and Srivastava, 2024), social enterprises struggle to

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manage AI-related uncertainties; hence, increasing their vulnerability to poor decisions in volatile environments. Missteps in AI adoption can have irreversible consequences, undermining mission integrity, long-term success, and competitiveness (Peprah and Ayaa, 2022). Yet, despite AI's growing role in entrepreneurship, its impact on social ventures remains understudied and scarce (Colombelli *et al.*, 2023). Without such insights, these entities risk misaligned strategies, wasted resources, and eroded market positioning in an increasingly AI-driven landscape.

The practical challenges of AI adoption further highlight this gap. In practice, it has been shown that while AI offers transformative potential, its implementation in social enterprises often creates tensions in decision-making, resource allocation, and mission alignment (Horosowski, 2023). Therefore, by systematically and empirically analyzing these dynamics, this research seeks to bridge the divide by examining *“how AI's unique capabilities influence decision-making in social entrepreneurship, with competitive advantage as an outcome and resource allocation as a moderator”*. By doing so, it aims to align theoretical promise with practical implementation in mission-driven contexts.

Therefore, this research leveraged the next-generation perceived characteristics of innovations (PCI) framework and the resource-based view (RBV) to explore its research question, collecting data ( $n = 269$ ) via e-surveys from different data sources: a business database and two AI-social entrepreneurship seminars. A quantitative methodology was employed, addressing the limited use of such designs in the social entrepreneurship literature (Cardella *et al.*, 2021). Notably, the regression analysis revealed the presence of nonlinear and moderating relationships; thus, offering insights to guide more tailored decision-making (Kao *et al.*, 2019).

This research addresses key research gaps by making four contributions across empirical, conceptual, theoretical, and practical dimensions. Empirically, it pioneers in examining how different AI capabilities, manifested through unique characteristics (Anagnostou *et al.*, 2022), influence decision-making; thus, addressing the research gap identified by Colombelli *et al.* (2023) and Neirotkh *et al.* (2024) regarding AI's understudied impact on social entrepreneurship and its implications for strategic decision-making. Conceptually, it introduces and develops a short measure of *“abstraction”* as a novel AI capability; hence, responding to urgent calls by Perifanis and Kitsios (2023) for introducing new AI capabilities to the literature. Theoretically, this research merges the next-generation PCI framework with RBV for a novel theoretical lens; therefore, addressing the undertheorized integration of AI and resource allocation in social entrepreneurship (Day *et al.*, 2025; Elgendy *et al.*, 2022). Practically, it develops a calibration graph as a *“prototype”* tool to quantify AI abstraction for resource-limited social entrepreneurship; thus, addressing urgent calls for advancing innovative solutions in social entrepreneurship (Goel, 2025).

This research follows a structured approach, starting with a thorough review of relevant literature, followed by a detailed analysis of the research methodology. The findings are then discussed, highlighting key insights. Finally, the implications of the results are explored, along with suggestions for future research directions.

## 2. Literature review

### 2.1 Artificial intelligence

The definition of AI remains ambiguous, with no universally accepted definition in literature (Kong *et al.*, 2023; Schuett, 2023). Various scholars offer different perspectives on AI, emphasizing its multifaceted nature. For instance, Mikalef and Gupta (2021), Makarius *et al.* (2020), and Schmidt *et al.* (2020) highlight AI's ability to interpret data and emulate human cognitive functions. In contrast, Demlehner and Laumer (2020) focus on AI's capacity to learn and adapt without explicit programming, while Wamba-Taguimdje *et al.* (2020) describe it as a combination of theories and techniques for modeling intelligent behavior with minimal human intervention. Afiouni (2019) and Longoni *et al.* (2019) emphasize AI's role in executing tasks

that typically require human intelligence. Additionally, [Rai et al. \(2019\)](#) and [Duan et al. \(2019\)](#) underline AI's capacity for reasoning and problem-solving, akin to human cognitive functions. Furthermore, AI encompasses not only machine learning, as noted by [Goodfellow et al. \(2016\)](#), but also robotics, as indicated by the [International Federation of Robotics \(2016\)](#). These diverse perspectives contribute to a comprehensive understanding of AI, which ranges from mimicking human cognition to autonomous operation for enhancing human well-being.

However, the definition of AI proposed by [Berente et al. \(2021\)](#) is adopted in this research. According to their definition, AI is characterized as “*the frontier of computational advancements that references human intelligence in addressing ever more complex decision-making problems*”. Therefore, this research adopts the definition of AI that emphasizes “*abstraction*” as an inherent characteristic of AI, which has not yet been empirically developed as a measured construct in the literature. It highlights AI's ability to analyze complex data and extract higher-level patterns for optimal decision-making.

In literature, AI has been classified into various techniques utilized by businesses, such as machine learning, rule-based modeling, fuzzy logic, visual processing, and natural language processing ([Sarker, 2022](#)). Nonetheless, this research specifically examines advanced analytics techniques, which encompass activities aimed at enhancing decision support systems to maximize competitive advantage ([Alazab et al., 2018](#)). Decision support systems are categorized as data-driven, communication-driven, document-driven, model-driven, or knowledge-driven ([Kumar, 2020](#)). However, this research does not differentiate among these categories due to the variability of their infrastructural components and objectives ([Soori et al., 2024](#)).

## 2.2 Social entrepreneurship

The literature has explored and expanded upon various types of entrepreneurship. On one hand, earlier literature identifies only two main types of entrepreneurship: general and social ([Mendez-Picazo et al., 2021](#)). General entrepreneurship focuses on profit-driven value creation through market opportunities, innovation, and new products or services. In contrast, social entrepreneurship prioritizes generating positive social impact through innovative business models ([Sanchez-Robles et al., 2023](#); [Tan Luc et al., 2022](#)).

On the other hand, recent literature has expanded the categorization of entrepreneurship to encompass various types, including social, corporate (intrapreneurship), digital/technological, environmental, and scalable startups ([Bhavsar, 2024](#); [Ratten, 2023](#)).

Furthermore, some studies have treated social startups as either equivalent to or as a subcategory of social entrepreneurship ([Waqar et al., 2020](#); [Fiofanova, 2020](#)). This research adopts a similar perspective, treating social entrepreneurship and social startups as conceptually, contextually, and methodologically equivalent. This approach is grounded in their shared social objectives (missions), commonalities in the application of AI decision systems, and their heavy reliance on limited resources ([Gupta and Srivastava, 2024](#)).

Consequently, despite that social entrepreneurship lacks a universally agreed-upon definition ([Ruiz-Rosa et al., 2020](#)), this research defines social entrepreneurship as “*the process by which individuals, startups and entrepreneurs develop and fund solutions that directly address social issues*” ([Peek, 2020](#)).

This definition is intentionally selected over broader alternatives to explicitly link social entrepreneurship to startups, reflecting their interdependent relationship and the significant role social entrepreneurship plays in startup management ([Barboulescu et al., 2021](#)), as also evident in this study's data collection targeting social entrepreneurship, social startups, users (individual employees), and entrepreneurs from different sources.

## 2.3 AI characteristics

Recent academic research has shifted from conventional technological attributes like usefulness and complexity, focusing instead on new AI-specific features ([Weber et al., 2023](#);

[Issa et al., 2022](#)). This shift highlights the inadequacy of traditional technological metrics in explaining the complexity of modern decision-making processes; thus, supporting claims that standard technologies alone are insufficient to capture the nuanced dynamics in today's markets ([Cao et al., 2021](#)). Therefore, this research investigates three established AI capabilities, which have been labeled as unique characteristics (i.e. interactivity, autonomy, and inscrutability), along with a newly proposed dimension, abstraction, as key antecedents to decision-making.

**2.3.1 Interactivity.** Interactivity is crucial in user-AI interactions, significantly influencing the quality and extent of engagement and serving as a fundamental component of AI system interfaces ([Du and Xie, 2021](#)). It is a vital aspect of technology-mediated environments ([Javornik, 2016](#)) and encompasses interactions among users, machines, and messages. Research has shown that interactivity is positively correlated with digitization ([Sundar et al., 2015](#)), advanced technologies like augmented and virtual realities ([Lakkis and Issa, 2022](#)), technology acceptance ([Pelau et al., 2021](#)), and AI readiness ([Issa et al., 2022](#)). In social entrepreneurship contexts, interactivity enhances user engagement with AI systems, facilitating better collaboration and communication in addressing complex societal issues.

**2.3.2 Autonomy.** Autonomy, initially understood as a personal trait, has evolved across multiple disciplines, including psychology and innovation. Earlier definitions of autonomy focused on independence and self-determination in decision-making ([Morgeson and Humphrey, 2006](#)). Today, autonomy is regarded as a key non-functional attribute of technological systems, reflecting their ability to adapt and adjust actions in response to environmental changes ([Janiesch et al., 2019](#)) with little to no human intervention ([Weber et al., 2023](#)). In entrepreneurship literature, autonomy has been shown to be the pillar for technology development (i.e. user-centric designs for communities), collaborative ecosystems (i.e. aligning missions with societies), and business models designs (introducing pricing strategies with social impact metrics) ([Thomson et al., 2022](#)).

**2.3.3 Inscrutability.** Inscrutability, or the complexity and opacity of AI systems, presents challenges for users who struggle to interpret their internal workings and probabilistic outcomes ([Johnk et al., 2021](#)). This characteristic can introduce uncertainty in AI project planning and deepen the knowledge gap between tech-focused and other business departments ([Weber et al., 2023](#)). However, in social entrepreneurship, inscrutability is harnessed to tackle complex societal challenges. AI empowers social entrepreneurs to tackle issues like poverty, healthcare, and sustainability with innovative approaches. Its inscrutability, while challenging, can also inspire creative solutions to these pressing problems.

**2.3.4 Abstraction.** Computational thinking, particularly abstraction as its central element, is vital for enhancing decision-making in mathematics by translating problems into mathematical forms ([Angraini et al., 2023](#)). In AI, abstraction enables systems to analyze complex data and derive high-level patterns or concepts, distinguishing AI from traditional technologies. This capability enhances strategic decision-making ([Korteling et al., 2021](#)); hence, benefiting social businesses by aligning decisions with their missions and increasing impact. AI's abstraction and related characteristics have become critical for entrepreneurial ventures, especially under uncertainty ([Townsend et al., 2023](#)).

### 3. Research model and hypotheses development

This research integrates the next-generation PCI framework and the RBV to examine ten hypothesized relationships (see [Figure 1](#)). The PCI framework, introduced by [Issa et al. \(2022\)](#), shifts focus from traditional technology characteristics (v relative advantage, trialability, usefulness, compatibility, etc.) ([Chatterjee et al., 2022](#)) to AI-specific traits like autonomy and interactivity, better reflecting today's dynamic technological advancements. Unlike older, but significant, models (e.g. technological, organizational and environmental and technology task fit), this approach captures AI's evolving role. Supporting this argument, a recent study in decision-making literature argues that current theoretical models must be

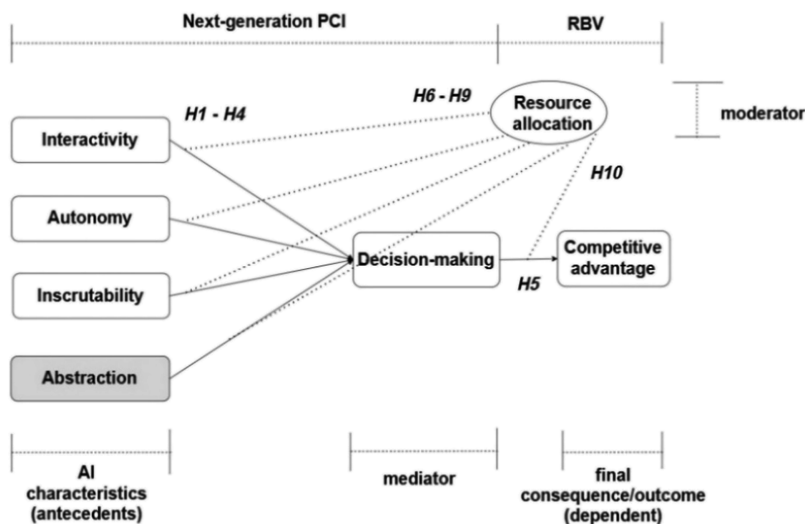


Figure 1. Hypothesized research model. Source: Authors' own work

revised or adapted to account for emerging technologies and advancements (i.e. [Elgendy et al., 2022](#)).

Complementing this, RBV ([Donnellan and Rutledge, 2019](#)) explains how AI traits become strategic resources when properly allocated, fostering competitive advantage ([Wamba-Taguimdje et al., 2020](#); [Reynolds, 2024](#)). Effective resource allocation ([Chen et al., 2021](#)), which has been conceptually proven to be a crucial element for social entrepreneurship ([Geradts and Alt, 2022](#)), positions AI as a key driver of competitiveness, with decision-making mediating this relationship. The merger of next-generation PCI and RBV integrates AI-specific features (e.g. interactivity, autonomy) with strategic resource management. By combining PCI's dynamic AI capabilities with RBV's resource focus, this approach demonstrates how AI's unique traits form strategic assets. The synergy improves the empirical model, showing how effective decision-making turns AI into sustainable competitive advantages.

### 3.1 AI characteristics – decision-making relationships

First, research indicates that interactivity enhances decision-making by improving transparency and comprehensibility ([Heleno et al., 2019](#)). It is conceptually recognized as a core feature in methods aimed at boosting decision-making effectiveness ([Hu et al., 2019](#)). Empirical evidence supports this, showing a positive correlation between interactivity and decision quality ([Tang, 2020](#)). Another experimental study further highlights that interactive visual tools significantly improve decision quality ([Samek et al., 2016](#)).

Second, autonomy has been shown in psychological studies to reduce the cognitive load required for decision-making by activating brain regions associated with working memory and logical reasoning. This enhancement supports more effective decision-making in human-technology collaborations ([Palmer et al., 2022](#)). A significant body of research highlights the positive impact of autonomy on decision-making. For example, studies in clinical environments suggest that a strong preference for autonomy in decision-making correlates with improved quality of life outcomes ([Karamched et al., 2018](#)). Likewise, findings from other fields consistently link autonomy with more efficient decision-making processes ([Fehr, 2015](#)).

Third, the characteristic of inscrutability, marked by opaqueness and ambiguity, is a common concern in the literature regarding AI systems. As noted by [Fainman \(2020\)](#), AI inherently lacks transparency in its decision-making processes. This opacity is a fundamental trait of algorithms, making them difficult to interpret ([Holzinger et al., 2020](#)) and often leading to questionable or unacceptable outcomes ([Chesterman, 2021](#)). Across various fields such as healthcare, management, and ethics, research has demonstrated that this ambiguity can hinder the effectiveness of decision-making ([Sun et al., 2020](#)).

Fourth, recent research has underscored the substantial influence of AI's computational capabilities on the effectiveness of decision-making ([Saha et al., 2023](#); [Wang, 2021a](#)), particularly in uncertain environments ([Galaz et al., 2021](#)). AI algorithms rely on their computational resources to inform decisions ([Mahmud et al., 2022](#)). Empirical studies have demonstrated that these computational algorithms positively relate to decision accuracy ([Vanderborght, 2019](#)). Therefore, based on the evidence from the literature outlined above, this research proposes the following hypotheses:

- H1. Interactivity as an AI characteristic positively relates to decision-making.
- H2. Autonomy as an AI characteristic positively relates to decision-making.
- H3. Inscrutability as an AI characteristic negatively relates to decision-making.
- H4. Abstraction as a novel AI characteristic positively relates to decision-making.

### 3.2 Decision-making and competitive advantage relationship

Effective decision-making is essential for achieving a competitive advantage and establishing a lasting position in the market ([Filip et al., 2019](#)). In technology research, digital tools have been shown to facilitate knowledge sharing and improve decision-making within organizations, strengthening their competitive position ([Deng et al., 2023](#)). Studies also reveal that sound decision-making supports competitive edge by helping organizations navigate complex market dynamics and capitalize on opportunities ([Calafut et al., 2023](#)). Furthermore, an empirical study by [Cao et al. \(2019\)](#) demonstrated that effective decision-making can act as a positive mediator between technological capabilities and competitive advantage, which is a finding that aligns with the hypothesis of this research. In addition, another recent study by [Dixit et al. \(2021\)](#) further demonstrated that strategic thinking, which is expressed as a decision-making process, serves as a key driver of competitive advantage. Hence, based on the evidence found in literature, it is plausible to hypothesize the following:

- H5. Effective decision-making positively relates to competitive advantage.

### 3.3 Resource allocation as a moderator

In technology and decision management research streams, resource allocation emerges as a key factor that ensures the optimal exploitation of limited resources ([Zhuang and Huang, 2019](#)). Research indicates that resource allocation not only moderates the link between technological innovation and decision-making ([Fragoulakis et al., 2017](#)) but also plays a crucial moderating role in both algorithmic decision processes ([Guo et al., 2019](#)) and the achievement of competitive advantage ([Panjaitan, 2022](#)). Specifically, resource commitment, which is closely tied to resource allocation, has been shown to be a significant moderator for technological resources as antecedents (e.g. [Mao et al., 2016](#)). In addition, prior research has established resource allocation as a significant moderator of organizational performance (e.g. [Attia, 2015](#)). Expanding on this, [Adomako and Ahsan \(2022\)](#) demonstrated that resource availability, when effectively managed through proper allocation, also plays a critical moderating role in entrepreneurial performance. Therefore, based on evidence found in literature, it is reasonable to hypothesize the following:



- H6-H9. Resource allocation positively moderates all AI characteristics and decision-making relationships.
- H10. Resource allocation positively moderates decision-making and competitive advantage relationship.

#### 4. Methodology

To efficiently gather responses ( $n = 269$ ) from participants with expertise in AI decision systems and social impact, data was collected from a business database, a conference, and a summit (see Figure 2 and Appendix A). A concise e-survey was used to test ten hypotheses while ensuring high response rates. The survey included six demographic variables (age, gender, education, employment status, AI experience, and social entrepreneurship background) (see Table 1) to ensure diversity. Additionally, 21 items measuring seven factors used validated scales with acceptable Cronbach alpha values (see Table 2). To avoid multicollinearity, demographics and independent variables were not used as controls (Shaheen et al., 2023).

##### 4.1 Abstraction: short scale development and convergence

Empirically, this research develops a concise three-phase short scale to measure “abstraction” as a novel AI capability (see Figure 3), which is conceptually characterized by its capacity to process complex datasets and detect higher-order patterns; hence, facilitating more sophisticated decision-making (Holmen and Saemundsson, 2021). Short-scale development, which focuses on creating concise measurement for existing or new

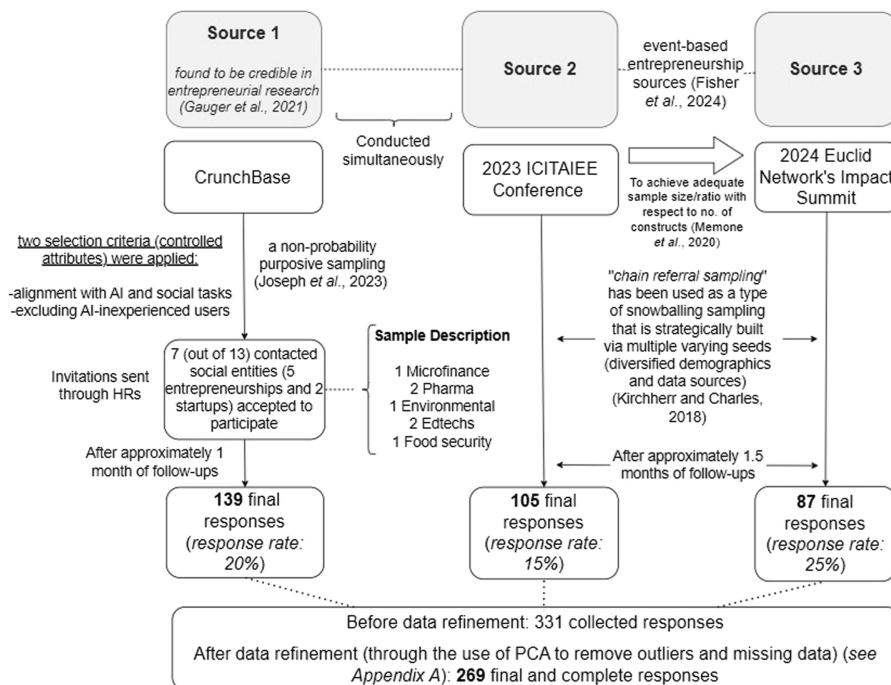


Figure 2. Data collection scheme. Source: Authors' own work

Table 1. Demographics

| Measures                                                    | Descriptions                    | Frequency | %     |
|-------------------------------------------------------------|---------------------------------|-----------|-------|
| Age                                                         | 18–26                           | 39        | 14.50 |
|                                                             | 27–37                           | 86        | 32    |
|                                                             | 38–48                           | 93        | 34.50 |
|                                                             | 49–59                           | 42        | 15.60 |
|                                                             | >60                             | 09        | 3.40  |
| Gender                                                      | Male                            | 65        | 24.20 |
|                                                             | Female                          | 204       | 75.80 |
| Education level                                             | Bachelors                       | 112       | 41.60 |
|                                                             | Masters                         | 124       | 46.10 |
|                                                             | Doctorate                       | 06        | 2.21  |
|                                                             | Others                          | 27        | 10    |
| Employment type                                             | Direct Employee (fixed or temp) | 155       | 57.60 |
|                                                             | Consultant                      | 42        | 15.60 |
|                                                             | Contractor                      | 06        | 2.20  |
|                                                             | Researcher                      | 15        | 5.60  |
|                                                             | Advocate                        | 32        | 11.90 |
|                                                             | Other                           | 19        | 7.10  |
| AI experience (related to advanced decision support system) | Less than 1 year                | 118       | 43.80 |
|                                                             | 1–3 years                       | 151       | 56.20 |
|                                                             | 4–6 years                       | –         | –     |
|                                                             | Above 6 years                   | –         | –     |
| Work experience (in social startups/entrepreneurships)      | Less than 1 year                | 54        | 20    |
|                                                             | 1–5 years                       | 132       | 49.10 |
|                                                             | 6–10 years                      | 83        | 30.90 |
|                                                             | 11–15 years                     | –         | –     |
|                                                             | Above 15 years                  | –         | –     |
| Total                                                       |                                 | 269       | 100   |

**Note(s):** Empirical evidence for the chosen demographics: Education may influence AI perceptions (Staikova et al., 2024), employment status could affect resource allocation (Lam, 2021), and AI experience may shape understanding of competitive advantages (Trawnih et al., 2022)

constructs, is highly reliable, as it builds on established constructs (Crede et al., 2012) and has proven to be both efficient and psychometrically sound (see Appendix B) (Novotny et al., 2025; Issa et al., 2024).

Conceptually, this novel AI capability is developed through design and functionality principles (see Figure 4). The design layer is anchored by three main AI pillars related to decision-making (i.e. infrastructural, architectural, and application) (Issa et al., 2023), in which each corresponding to distinct AI functionalities identified in prior literature (Bakir and Bakir, 2025; Swan et al., 2023; Liu et al., 2020). These functionalities, including computational thinking, dynamism, and responsiveness, have been empirically validated and operationalized as short-scale parameters for measuring abstraction (Issa, 2022; Korkmaz et al., 2017; Pandit et al., 2017; Pavlou and El Sawy, 2011) (see Table 2).

5. Results

Three tools were used for the statistical analysis and validity tests (SPSS v.26, AMOS v. 29, and STATA v. 14.2). First, common method bias was assessed and validated using Harman’s One-Factor Test (SPSS) and Principal Component Analysis (STATA). Both tests provided adequate estimates; thus, indicating and confirming that common method bias was not a significant issue (see Appendix C). Second, the factor loadings showed significant correlations (Hair et al., 2006; Podsakoff et al., 2003; Chin, 1998) (see Table 2), and the validity



**Table 2.** Questionnaire items

| Item code                                                                                                                                                                                                                                                       | Item description                                                                                                                                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Interactivity</i> (Lakkis and Issa, 2022; Pillai <i>et al.</i> , 2020)                                                                                                                                                                                       |                                                                                                                                                      |
| <i>Cronbach alpha:</i> 0.812; <i>Factor loadings:</i> 0.777; 0.850; 0.862                                                                                                                                                                                       |                                                                                                                                                      |
| INT 1                                                                                                                                                                                                                                                           | I felt that I had a lot of control over my experiences with the advanced decision-support system                                                     |
| INT 2                                                                                                                                                                                                                                                           | When using the advanced decision-support system, I felt I was getting instantaneous information                                                      |
| INT 3                                                                                                                                                                                                                                                           | Using the advanced decision-support system is very engaging                                                                                          |
| <i>Autonomy</i> (Issa <i>et al.</i> , 2022; Morgeson and Humphrey, 2006)                                                                                                                                                                                        |                                                                                                                                                      |
| <i>Cronbach alpha:</i> 0.771; <i>Factor loadings:</i> 0.865; 0.866; 0.718                                                                                                                                                                                       |                                                                                                                                                      |
| AUT 1                                                                                                                                                                                                                                                           | The advanced decision-support system is effectively able to choose its own method to develop applications and finish tasks                           |
| AUT 2                                                                                                                                                                                                                                                           | The advanced decision-support system has full control over which type of applications to design or tasks to complete                                 |
| AUT 3                                                                                                                                                                                                                                                           | The advanced decision-support system is fully capable to decide on its own what/which applications should be designed or tasks to finish first       |
| <i>Inscrutability</i> (adapted from IT ambiguity (Pillet <i>et al.</i> , 2018))                                                                                                                                                                                 |                                                                                                                                                      |
| <i>Cronbach alpha:</i> 0.889; <i>Factor loadings:</i> 0.901; 0.884; 0.872                                                                                                                                                                                       |                                                                                                                                                      |
| INS 1                                                                                                                                                                                                                                                           | In general, it is not clear what the advanced decision-support system is and is not                                                                  |
| INS 2                                                                                                                                                                                                                                                           | Overall, I find it difficult to make sense of the advanced decision-support system                                                                   |
| INS 3                                                                                                                                                                                                                                                           | All in all, the advanced decision-support system is hard to fully comprehend                                                                         |
| <i>Abstraction</i> (adapted from computational/algorithmic thinking (Korkmaz <i>et al.</i> , 2017) and dynamic capability of innovation (Pandit <i>et al.</i> , 2017; Pavlou and El Sawy, 2011) and responsiveness characteristic of technologies (Issa, 2022)) |                                                                                                                                                      |
| <i>Cronbach alpha:</i> 0.846; <i>Factor loadings:</i> 0.849; 0.868; 0.866                                                                                                                                                                                       |                                                                                                                                                      |
| ABS 1                                                                                                                                                                                                                                                           | The advanced decision-support system learns better the instructions made with the help of mathematical symbols and concepts                          |
| ABS 2                                                                                                                                                                                                                                                           | The advanced decision-support system is fully aware in responding to the external environment                                                        |
| ABS 3                                                                                                                                                                                                                                                           | The advanced decision-support system integrates individual knowledge and inculcates it into the operational capabilities via collective sense-making |
| <i>Resource-allocation</i> (Pulles <i>et al.</i> , 2016)                                                                                                                                                                                                        |                                                                                                                                                      |
| <i>Cronbach alpha:</i> 0.822; <i>Factor loadings:</i> 0.812; 0.872; 0.864                                                                                                                                                                                       |                                                                                                                                                      |
| RES 1                                                                                                                                                                                                                                                           | My entrepreneurship allocates more financial resources to the relationship with the advanced decision-support system                                 |
| RES 2                                                                                                                                                                                                                                                           | In case of capacity bottlenecks, my entrepreneurship allocates our scarce resources to the advanced decision-support system instead to others        |
| RES 3                                                                                                                                                                                                                                                           | My entrepreneurship allocates the best users/employees (e.g. most experienced, trained, intelligent) to manage the advanced decision-support system  |
| <i>Decision-making</i> (Zhang <i>et al.</i> , 2019)                                                                                                                                                                                                             |                                                                                                                                                      |
| <i>Cronbach alpha:</i> 0.842; <i>Factor loadings:</i> 0.784; 0.870; 0.856                                                                                                                                                                                       |                                                                                                                                                      |
| DEC 1                                                                                                                                                                                                                                                           | My entrepreneurship is able to make future plans of its goals through the use of the advanced decision-support system                                |
| DEC 2                                                                                                                                                                                                                                                           | My entrepreneurship is able to solve problems through the use of the advanced decision-support system                                                |
| DEC 3                                                                                                                                                                                                                                                           | My entrepreneurship is able to make accurate appraisals through the use of the advanced decision-support system                                      |
| <i>Competitive advantage</i> (Saeidi <i>et al.</i> , 2015)                                                                                                                                                                                                      |                                                                                                                                                      |
| <i>Cronbach alpha:</i> 0.905; <i>Factor loadings:</i> 0.841; 0.919; 0.887                                                                                                                                                                                       |                                                                                                                                                      |
| COM 1                                                                                                                                                                                                                                                           | My entrepreneurship is able to sustain a good growth because of the advanced decision-support system                                                 |
| COM 2                                                                                                                                                                                                                                                           | My entrepreneurship is able to gain a differentiation and diversity because of the advanced decision-support system                                  |
| COM 3                                                                                                                                                                                                                                                           | My entrepreneurship is able to achieve market leadership because of the advanced decision-support system                                             |

**Note(s):** Throughout the e-survey, the term “entrepreneurship” referred specifically to “social entrepreneurship.” To ensure respondents were from such organizations, the e-survey included a screening/filter question confirming their affiliation with a social enterprise

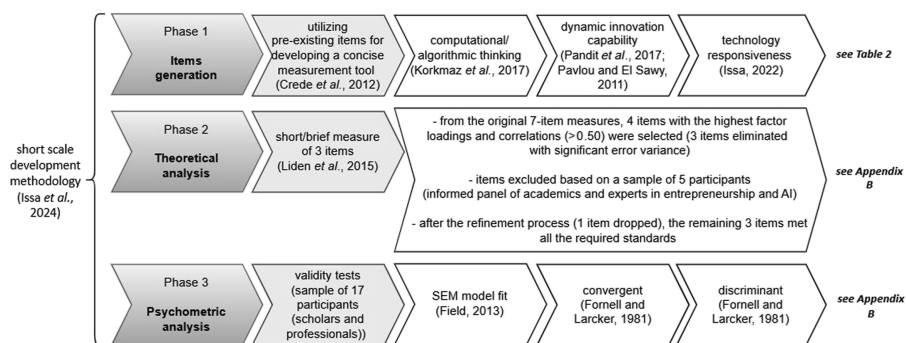


Figure 3. Tri-phasic short scale development. Source: Authors' own work

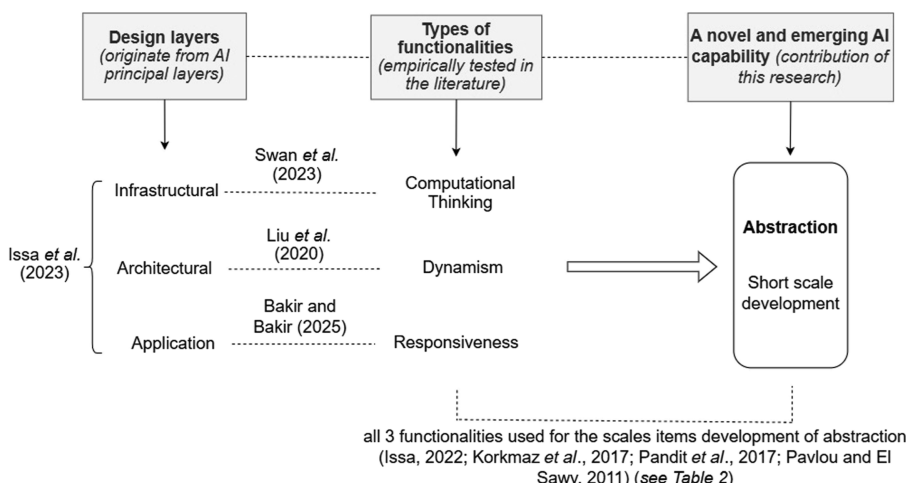


Figure 4. Convergence of abstraction. Source: Authors' own work

(convergent and discriminant) showed satisfactory estimates (Fornell and Larcker, 1981) (see Appendix C). Furthermore, through conducting structural equation modeling, the appropriateness of the model and robustness validity were assessed; thus, both providing adequate measurement model fit and showed no multicollinearity concerns (see Table 3).

For empirical insights, the hierarchical regression analysis within a frequentist framework (Wang, 2021b) yielded intriguing results. Notably, age was the only control variable that showed a significant relationship with decision-making and competitive advantage (see Appendix C). In terms of the framework, three hypotheses (H1, H6, and H7) lacked support, while the others were upheld (see Table 3). Specifically, resource-allocation did not act as a moderator for the relationships between interactivity, autonomy characteristics, and decision-making (see Appendix D for all moderation graphs). However, despite the lack of a significant linear correlation between interactivity and decision-making, nonlinear regression analysis revealed a cubic (s-shaped) relationship (also see Table 3). Initially, greater interactivity improves decision quality by fostering diverse insights, innovation, and alignment. However, excessive interactivity can hinder decisions due to conflicts, paralysis, or diluted focus. Beyond this point, decision-making

**Table 3.** Hierarchical regression analysis (ordinary least squares (OLS))

| Variables                                                                                                                                                                                                                                                                                                           | <i>t</i>    | Std $\beta$   | Unst. $\beta$            | Hypotheses                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------------|--------------------------|---------------------------|
| Model 1 ( <i>four antecedents that predict the mediator</i> )                                                                                                                                                                                                                                                       |             |               |                          |                           |
| Decision-making (DEC)                                                                                                                                                                                                                                                                                               |             |               |                          |                           |
| Interactivity (INT)                                                                                                                                                                                                                                                                                                 | −0.767      | −0.048        | −0.051                   | <i>H1 not supported</i>   |
| Automation (AUT)                                                                                                                                                                                                                                                                                                    | 2.058**     | 0.121**       | 0.178**                  | <i>H2 supported</i>       |
| Inscrutability (INS)                                                                                                                                                                                                                                                                                                | −2.067**    | −0.130**      | −0.123**                 | <i>H3 supported</i>       |
| Abstraction (ABS)                                                                                                                                                                                                                                                                                                   | 4.351*      | 0.255*        | 0.260*                   | <i>H4 supported</i>       |
| Resource Allocation (RES)                                                                                                                                                                                                                                                                                           | 2.659*      | 0.161*        | 0.189*                   | (supported)               |
|                                                                                                                                                                                                                                                                                                                     | $R = 0.325$ | $R^2 = 0.106$ | Adjusted $R^2 = 0.092$   |                           |
| Model 2 ( <i>mediator that predicts the outcome</i> )                                                                                                                                                                                                                                                               |             |               |                          |                           |
| Competitive advantage (COM)                                                                                                                                                                                                                                                                                         |             |               |                          |                           |
| Decision-making (DEC)                                                                                                                                                                                                                                                                                               | 7.811*      | 0.431*        | 0.439*                   | <i>H5 supported</i>       |
| Resource Allocation (RES)                                                                                                                                                                                                                                                                                           | 1.884***    | 0.115***      | 0.137***                 | (supported)               |
|                                                                                                                                                                                                                                                                                                                     | $R = 0.431$ | $R^2 = 0.186$ | Adjusted $R^2 = 0.183$   |                           |
| Model 3 ( <i>moderator that interacts with each antecedent and the mediator</i> ) and ( <i>moderator that interacts with the mediator and the outcome</i> )                                                                                                                                                         |             |               |                          |                           |
| Moderator 1 (RES $\times$ INT with DEC)                                                                                                                                                                                                                                                                             | −0.681      | −0.052        | −0.012                   | <i>H6 not supported</i>   |
| Moderator 2 (RES $\times$ AUT with DEC)                                                                                                                                                                                                                                                                             | 1.576       | 0.118         | 0.028                    | <i>H7 not supported</i>   |
| Moderator 3 (RES $\times$ INS with DEC)                                                                                                                                                                                                                                                                             | −1.977**    | −0.143**      | −0.029**                 | <i>H8 supported</i>       |
| Moderator 4 (RES $\times$ ABS with DEC)                                                                                                                                                                                                                                                                             | 4.705*      | 0.313*        | 0.071*                   | <i>H9 supported</i>       |
| Moderator 5 (RES $\times$ DEC with COM)                                                                                                                                                                                                                                                                             | 6.973*      | 0.392*        | 0.079*                   | <i>H10 supported</i>      |
| Model fit (SEM)                                                                                                                                                                                                                                                                                                     |             |               |                          |                           |
| CMIN/DF = 1.271 (<3)                                                                                                                                                                                                                                                                                                |             |               |                          |                           |
| CFI = 0.983 (>0.900)                                                                                                                                                                                                                                                                                                |             |               |                          |                           |
| RMSEA = 0.032 (<0.080)                                                                                                                                                                                                                                                                                              |             |               |                          |                           |
| <a href="#">Hair et al. (2010)</a> , <a href="#">Hooper et al. (2008)</a>                                                                                                                                                                                                                                           |             |               |                          |                           |
| Robustness validity ( <i>for Models 1 and 2</i> )                                                                                                                                                                                                                                                                   |             |               |                          |                           |
| VIF estimate for sub-model 1 = $1/1 - R^2 = 1.12$ ****                                                                                                                                                                                                                                                              |             |               |                          |                           |
| VIF estimate for sub-model 2 = $1/1 - R^2 = 1.23$ ****                                                                                                                                                                                                                                                              |             |               |                          |                           |
| <a href="#">Vaithilingam et al. (2024)</a> , <a href="#">Gruber et al. (2010)</a>                                                                                                                                                                                                                                   |             |               |                          |                           |
| Model 4 ( <i>Nonlinear relationship between INT and DEC</i> )                                                                                                                                                                                                                                                       |             |               |                          |                           |
| INT (1st slope; positive)                                                                                                                                                                                                                                                                                           | 2.049**     | 2.465**       | 2.581**                  | Nonlinear<br>significance |
| INT <sup>2</sup> (2nd slope; negative)                                                                                                                                                                                                                                                                              | −2.040**    | −5.609**      | −0.912**                 |                           |
| INT <sup>3</sup> (3rd slope; positive)                                                                                                                                                                                                                                                                              | 1.949**     | 3.140**       | 0.094**                  |                           |
|                                                                                                                                                                                                                                                                                                                     | $R = 0.155$ | $R^2 = 0.024$ | Adjusted $R^2 = 0.013$ ; |                           |
| <b>Note(s):</b> SEM = structural equation model; CMIN/DF = Chi-square minimum discrepancy divided by degrees of freedom; CFI = comparative fit index; RMSEA = root mean squared error of approximation; VIF = variance inflation factor; * < 0.01; ** < 0.05; *** < 0.06; **** < 5 (multicollinearity not an issue) |             |               |                          |                           |

improves again with structured input management, selective engagement, and advanced decision-support tools. These findings suggest that optimal (not maximal) interactivity is key for effective decision-making, reinforcing that “more” is not always “better.” This nuanced insight complements recent studies on how AI influences entrepreneurial decision-making (e.g. [Julie et al., 2024](#); [Nadi et al., 2024](#)).

## 6. Discussion

This research makes few modest, but significant, contributions. Empirically, it pioneers in examining how different AI capabilities, manifested through unique characteristics

(Anagnostou *et al.*, 2022), influence decision-making; thus, addressing the research gap identified by Colombelli *et al.* (2023) and Neirotkh *et al.* (2024) regarding AI's understudied impact on social entrepreneurship and its implications for strategic decision-making. Conceptually, it introduces and develops a short measure of "abstraction" as a novel AI capability; hence, responding to urgent calls by Perifanis and Kitsios (2023) for introducing new AI capabilities to the literature. Theoretically, this research merges the next-generation PCI framework with RBV for a novel theoretical lens; therefore, addressing the undertheorized integration of AI and resource allocation in social entrepreneurship (Day *et al.*, 2025; Elgendy *et al.*, 2022). Practically, it develops a calibration graph as a "prototype" tool to quantify AI abstraction; thus, addressing urgent calls for advancing innovative solutions in social entrepreneurship (Goel, 2025).

### 6.1 Empirical contribution

Empirically, this research pioneers in addressing the research gap identified by Colombelli *et al.* (2023) and Neirotkh *et al.* (2024) regarding AI's understudied impact on social entrepreneurship and its implications for strategic decision-making. Thus, given AI's inherent complexity and significant resource demands, which often pose challenges for social entrepreneurship (WEF, 2024), this empirical investigation provides timely and actionable insights. By systematically analyzing the positive and negative relationships among key AI capabilities as antecedents, outcomes, and a moderator, this research enables social entrepreneurs to make informed decisions about which AI characteristics (capabilities) deliver tangible value and which may drain limited resources. This issue has been evident in social entrepreneurship due to the mismatch among social businesses' goals and AI's own capabilities (e.g. automation, etc.) (Muttaki *et al.*, 2025). Therefore, moving beyond theoretical speculation, this research establishes an evidence-based investigation to guide AI use in social entrepreneurship, ensuring that strategic choices are grounded in rigorous empirical analysis rather than assumptions or trial-and-error.

### 6.2 Conceptual contribution

Conceptually, this research pioneers the introduction and development of a concise measure of "abstraction" as a novel AI capability, thereby responding to urgent calls by Perifanis and Kitsios (2023) for expanding the literature on AI capacities. Grounded in theory and empirically tested, this research defines abstraction as "the ability of AI to detect patterns, adapt dynamically, and respond contextually across infrastructural, architectural, and application layers". This contribution is critical because the AI tools most widely used by social entrepreneurship, such as machine learning (ML) and natural language processing (NLP), fundamentally depend on detecting patterns in data (WEF, 2024). As a core capability for both ML and NLP, abstraction enables this advanced pattern recognition, a concept formally operationalized in this research. For instance, in socially driven contexts, abstraction empowers AI to detect poverty patterns in specific areas, adapt outreach strategies in real time, and predict trends by integrating satellite data (infrastructural), ML models (architectural), and policy decisions (application).

### 6.3 Theoretical contribution

Despite the growing adoption of AI in social entrepreneurship, its integration remains largely undertheorized, often lacking robust theoretical grounding (Elgendy *et al.*, 2022). This has led to mismatched solutions that struggle to address real-world complexities. Furthermore, while the RBV has been widely applied in strategic management, its intersection with AI, particularly in the context of social entrepreneurship, remains underexplored (Day *et al.*, 2025). Consequently, this research addresses these gaps by merging the next-generation PCI framework with RBV; thus, offering a novel theoretical lens to examine AI's role in social

entrepreneurship. By integrating PCI's AI-centric traits with RBV's strategic resource allocation principles, this research demonstrates how AI's unique capabilities translate into sustainable competitive advantages. This synthesis not only advances AI theoretical models but also extends RBV by positioning AI as a critical strategic resource, mediated by effective decision-making. Hence, unlike prior studies that treat AI as a generic tool, such a novel theoretical lens redefines AI as a dynamic and unique resource, rather than a static asset, that enhances decision-making and consequently competitive advantage.

#### 6.4 Practical contributions

Practically, this research introduces a calibration graph (see Figure 5) as a "prototype" tool, rather than a definitive resolution, to quantify AI abstraction; thus, addressing urgent calls for advancing innovative solutions in social entrepreneurship (Goel, 2025). This tool helps social entrepreneurs precisely determine the optimal level of AI abstraction; thus, avoiding the risks of underestimation (failing to generalize solutions) or overestimation (applying overly complex models beyond available computing power). Therefore, by providing an actionable, data-driven method, rather than relying on theoretical managerial interventions, this tool empowers social entrepreneurship to strategically align their AI capabilities with their available resources, which consequently may lead to competitive advantage (Day et al., 2025).

Hence, the calibration graph translates these technical AI metrics (i.e. z-axis responsiveness (queries/second), y-axis dynamic capability (loss/error rate), and x-axis computational thinking (FLOPS)) (Ganbarov et al., 2024; Kowshik et al., 2021; Hernandez and Brown, 2020) into actionable insights for social entrepreneurship. For example, a health-focused

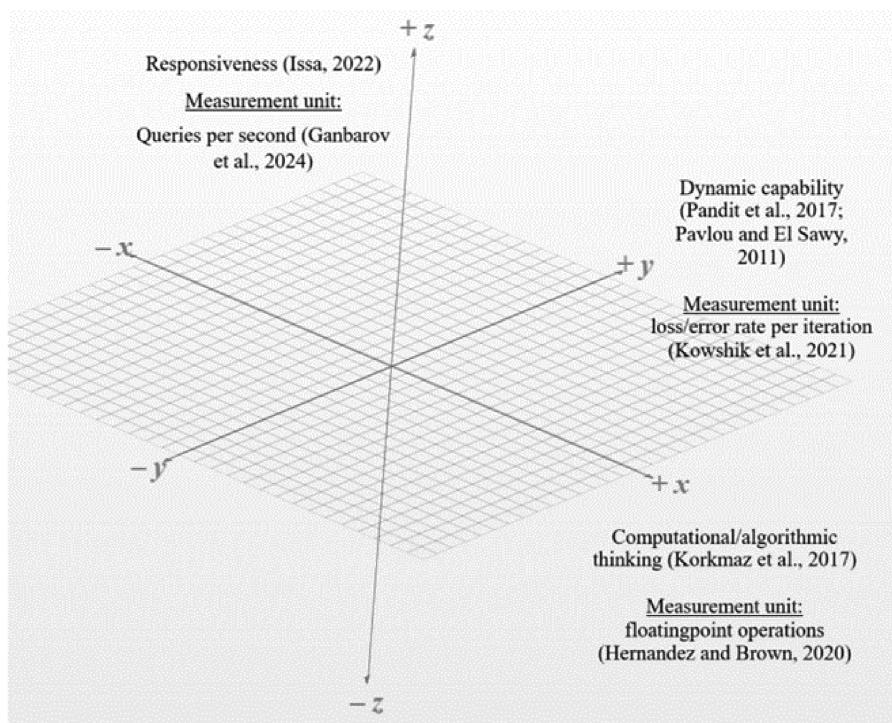


Figure 5. Calibration graph. Source: Authors' own work

social enterprise could use the graph to determine the optimal balance between AI responsiveness (e.g. real-time disease diagnostics in low-bandwidth areas) and error rates (e.g. ensuring algorithmic fairness in patient triage), while respecting hardware constraints (FLOPS). By plotting these metrics against resource limits (e.g. budget, data quality, or device capabilities), the graph helps social entrepreneurs visually identify trade-offs—such as sacrificing some speed for higher accuracy in poverty-mapping algorithms—and avoid over-engineering solutions that exceed their computational capacity. This operational approach ensures AI adoption is practical, scalable, and mission-aligned, turning abstract performance metrics into decisions that directly enhance social impact (see [Appendix E](#) for full operationalization).

### 7. Limitations and future research directions

This research acknowledges limitations in three areas. First, the pilot studies had underpowered sample sizes for short-scale development (Phases 2–3), though they met the minimum 5-to-1 sample-to-item ratio (22 responses >15 required) ([Memone et al., 2020](#)). Future studies are encouraged to adopt the 15-to-1 sample-to-variable ratio ([Hair et al., 2018](#)). Second, while the main study addressed selection bias through demographic comparisons ([Tetleh et al., 2021](#)), future studies should implement the “Inverse Probability Weighting” ([Breen and Ermisch, 2024](#)) for more rigorous correction. Third, the findings have constrained generalizability due to the niche context of social entrepreneurship, which requires further validation of the abstraction scale. Future research should expand to broader organizational contexts to externally validate the abstraction scale, ensuring its applicability beyond social entrepreneurship. Additionally, studies should address the field’s inherent imbalance in AI resource distribution ([Mugunzva and Manchidi, 2024](#)), particularly between social entities and well-funded tech organizations.

### 8. Conclusion

This research advances the discourse on AI’s role in social entrepreneurship by uncovering how its distinct characteristics shape strategic decision-making and competitive advantage. By empirically analyzing interactivity, autonomy, inscrutability, and the newly introduced dimension of abstraction, this research provides a nuanced understanding of AI’s influence in resource-constrained, mission-driven contexts. This was achieved by the integration of the next-generation PCI framework with RBV, which offered a novel theoretical lens to bridge gaps in AI, decision-making, and resource management literature. Consequently, the development of a calibration tool equips social entrepreneurship with actionable insights to navigate AI abstraction while maintaining resource alignment. These contributions not only enhance academic discourse but also enable mission-driven entities to leverage AI for strategic decision-making; thus, helping them achieve their core objectives in a highly competitive landscape.

### Availability of data and material

The collected and analyzed dataset is available on request.

### Supplementary material

The supplementary material for this article can be found online.

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